

Paipu Genome Preparation

Pipeline Set Up

User is responsible for setting these elements up for the pipeline.

1. Download DEXSeq script
2. Replace paths/slurm information in the Nextflow pipeline (*run_genome_prep_slurm.sh*)
3. Create Input File EX:

Ailuropoda melanoleuca
Antilocapra americana
Aotus nancymae

Query Genome Information

Query reference genome information for each mammal in input file.

ncbi_queries/\${mammal}.json

Parse Genome Information

Parse queried genome information to find the most recent genome for each mammal and available gene annotation files.

output_all.tsv

output.tsv

Download Mammal Reference Genome

Download reference genome and gene annotation files identified by genome query for each mammal

genomic.fna

genomic.gtf

Prep Genome Indexes

Prep FAI index, sequence dictionary, and HISAT index

Prepare GFF File

dexseq_prepare_annotation.py

Prepare GFF files for DEXSeq

Genomes are ready for Paipu!